4.5 Solving Polynomial Equations

Name _____ Period ____ Date _____

Algebra II 4.5 (30-minute plan) · Student sheet

I can…

- solve polynomial equations and find the zeros of polynomial functions.
- use the Rational Root Theorem to find possible rational solutions.
- write a polynomial function from its zeros, including irrational conjugates.

Vocabulary · fill in the missing word

Repeated solution — a factor that _____ more than once, such as $(x - 3)^2$

Rational Root Theorem — every rational _____ of $a_n x^n + \ldots + a_0 = 0$ has the form $\pm(\text{factor of } a_0)/(\text{factor of } a_n)$

Irrational Conjugates Theorem — if $a + \sqrt{b}$ is a zero of a polynomial with rational _____, so is $a - \sqrt{b}$

Part 1 · Notes and examples (fill in each note, then work the example)

Example 1

Solve $2x^4 - 12x^3 + 18x^2 = 0$.

Rational Root Theorem: Possible rational zeros = ±(factors of _____) / (factors of _____)

Irrational Conjugates Theorem: If $a + \sqrt{b}$ is a zero, then _____ is also a zero.

Example 2

Find all real zeros of $f(x) = x^3 - 8x^2 + 11x + 20$.

Part 2 · Partner practice

Partner practice

Solve $x^3 + x^2 - 6x = 0$.